TECHNOLOGY

AWS Solution Architect

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TECHNOLOGY

Security and IAM

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Learning Objectives

By the end of this lesson, you will be able to:

- Illustrate and configure models in AWS for resource optimization and enhancing overall operational efficiency
- Define Identity and Access Management in AWS for enabling secure, efficient, and compliant resource management

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Responsibility Models in AWS

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Shared Responsibility Model

AWS and the customer share responsibility for security and compliance.

- This shared approach helps reduce the operational load on the customer.
- AWS manages the host operating system, virtualization layer, and physical security of the buildings.

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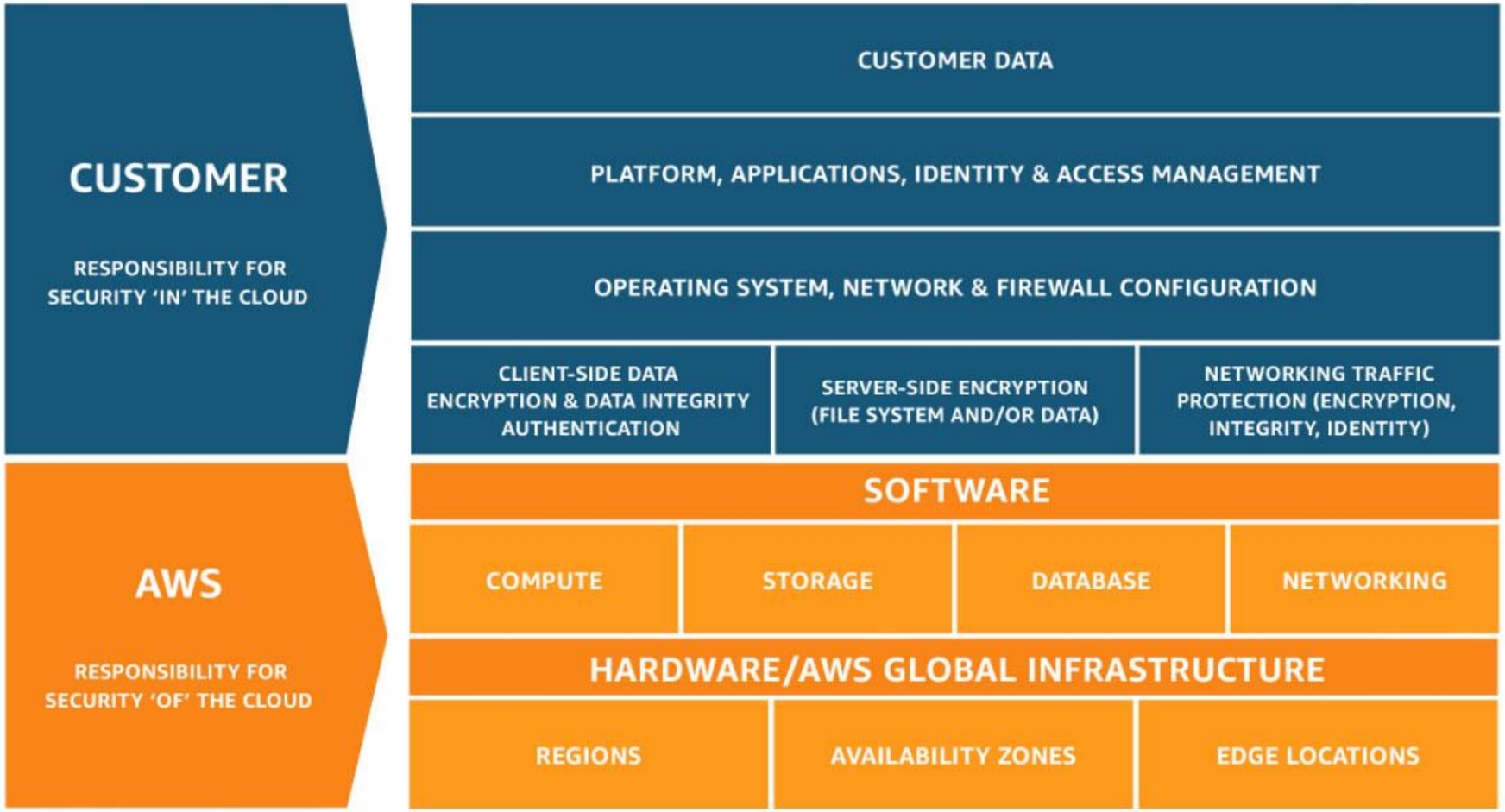
Shared Responsibility Model

Here is the working diagram of the AWS shared responsibility model:



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Compliance Program

AWS and clients share compliance responsibilities when systems are built in the AWS Cloud.

- Customers can delve into the comprehensive security measures maintained by AWS.
- AWS Compliance Enablers bolster traditional programs by integrating governance-focused, audit-friendly service features aligned with relevant compliance standards.

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AWS Artifacts

The following are the AWS artifacts:

- On-demand access to AWS
- Compliance reports
- Globally available
- Easy identification
- Quick assessments
- Continuous monitoring
- Enhanced transparency

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Delegation

It is a crucial aspect of managing access and control in AWS, and it helps organizations efficiently distribute responsibilities while maintaining security and compliance.

A registered member account is utilized for administrative AWS Single Sign-On (SSO) tasks.

The management account within AWS Organizations automatically generates an SSO instance.

The roles are easily managed across all member accounts within the organization using AWS.

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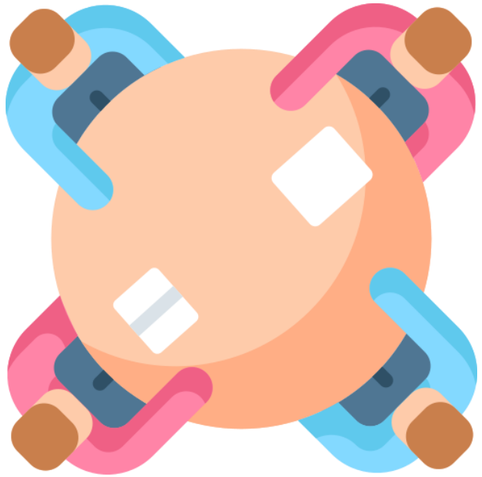
Benefits of Delegation

Delegation in AWS offers the following benefits:



It reduces the need for numerous individuals to access the management account, helping to mitigate security risks.

It empowers a limited set of administrators to assign users and groups to applications and user accounts for organization members.

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Federation

Identity federation is a trust-based system that connects two parties to authenticate users and transmit the necessary data to grant them access to resources.

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Enabling Federated AWS Access

- Through federation, users can access their AWS accounts using single sign-on (SSO).
- Users can utilize AWS IAM to control fine-grained, federated access to AWS accounts.
- Users can enable federated access for user-facing mobile and online applications.

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Quick Check

A company needs access to AWS compliance documents to understand their security responsibilities before migrating sensitive data. Which AWS service should they use?

A. AWS Shield

B. AWS Artifact

C. AWS CloudTrail

D. AWS IAM

AWS IAM

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Security Credentials

AWS offers a range of options to securely grant users access to their AWS resources, including:

Email address and password:
They are used for signing into the AWS Console.

IAM username and password:
They provide access to an AWS account for diverse users and programs.

Access keys:
Access keys are utilized to authorize programmatic requests.

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AWS Multi-Factor Authentication

Multi-factor authentication (MFA) adds a layer of security to the sign-in process.

It requires users to authenticate using an AWS-supported MFA method along with their sign-in credentials when accessing AWS services.

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AWS Multi-Factor Authentication

The following AWS-supported MFA mechanisms are available:

Virtual MFA Devices

This type of MFA employs an application on a phone or other device to generate a six-digit numeric code.

U2F Security Key

When a device is plugged into a USB port on the user's computer, the U2F security key serves as a form of MFA.

SMS Text Message-Based MFA

This MFA method involves adding the user's SMS-compatible cell phone number to the IAM user settings.

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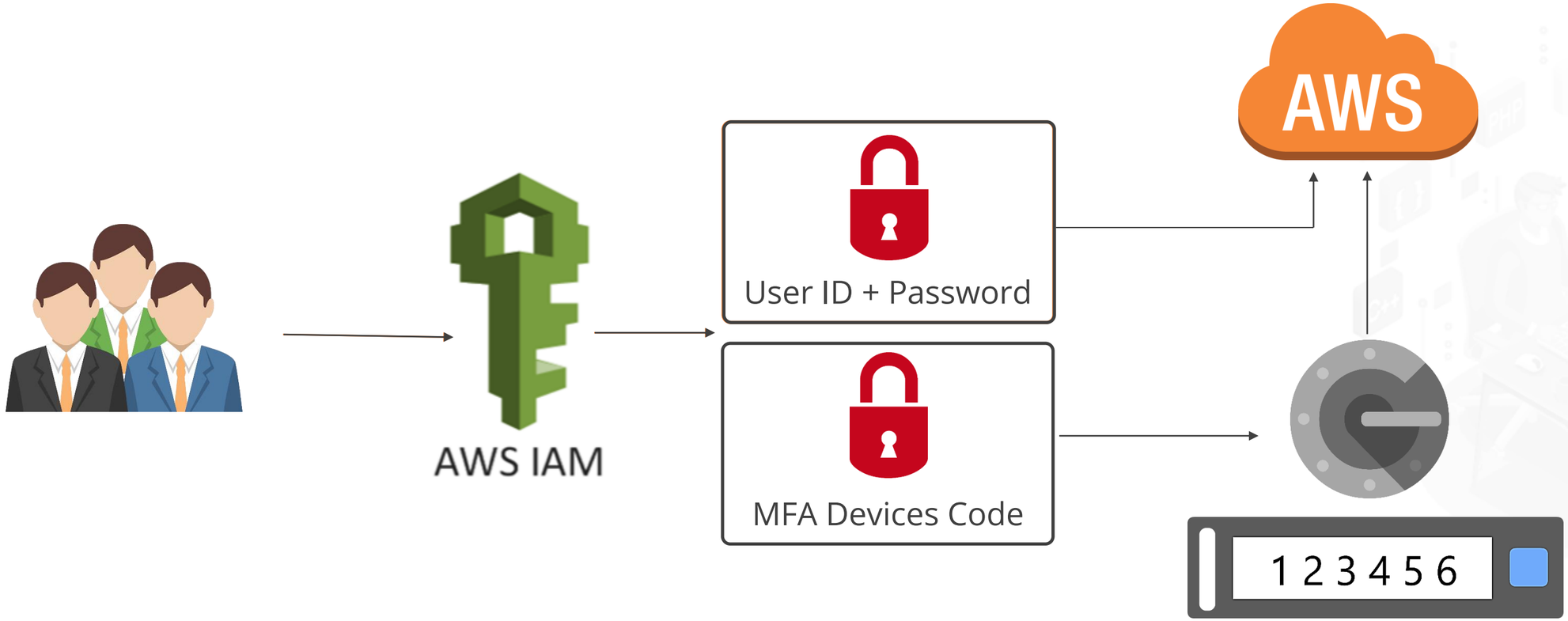
Multi-Factor Authentication

AWS IAM offers support for Multi-Factor Authentication (MFA), which ensures comprehensive security using MFA devices.



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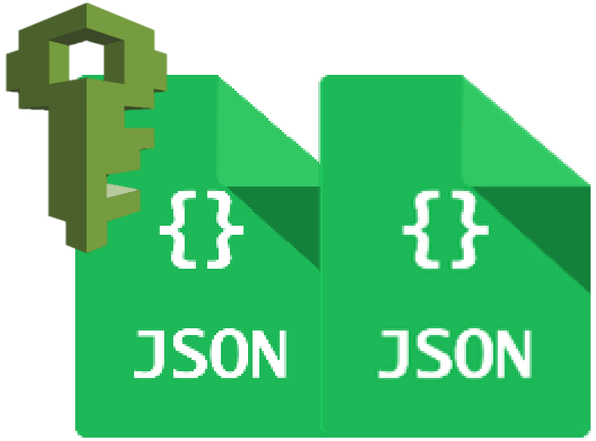
What Is IAM Policy?

An IAM policy is a document that defines individual or multiple permissions. These policies can be linked to users, groups, roles, and AWS resources, and are written in JSON format.



AWS IAM Policies

01 AWS provides a selection of predefined IAM Policies that users can choose from.

02 For increased customization, root users can modify these predefined policies.

03 Root users can create entirely new custom IAM policies from scratch.

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Types of IAM Policies

Two types of IAM policies exist:

Identity-based policies

These policies can be directly linked to identities like users, groups, and roles.

Resource-based policies

These policies are attached to AWS resources such as Amazon S3, Amazon EC2, and more.

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Syntax of Writing AWS IAM Policies

AWS policies are written using JavaScript Object Notation (JSON).

```
"Version":"2012-10-17",
  "Statement":[
    {
      "Sid":"statement1",
      "Effect":"Allow",
      "Action":"s3:ListAllMyBuckets",
      "Resource": "arn:aws:s3:::*"
  }
```

Policy-wide information:

Version: Date when the policy was created

One or more individual statements:

Effect: Allows permission
Action: Lists all the S3 buckets
Resource: Name of the S3 bucket

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Inline Policy

An IAM identity can contain an inline policy, which is a policy directly embedded within the identity (user, group, or role).

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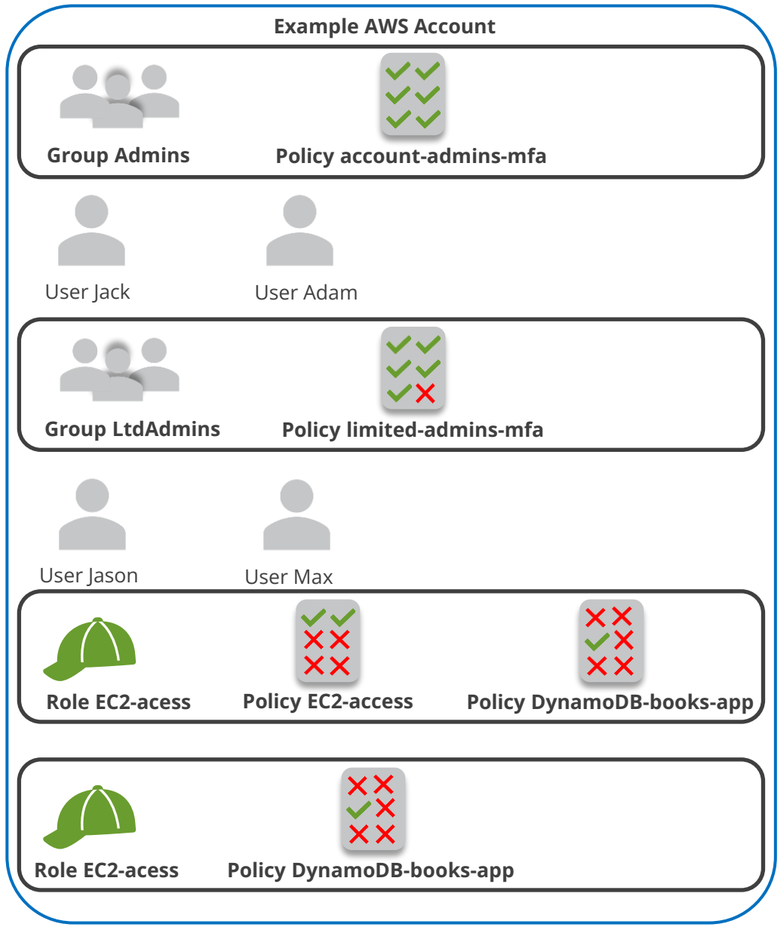
Inline Policy

The following image explains inline policies:



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Managed Policy

An AWS-managed policy is created and maintained by AWS as an independent policy.

Example:

```
arn:aws:iam::aws:policy/IAMReadOnlyAccess
```

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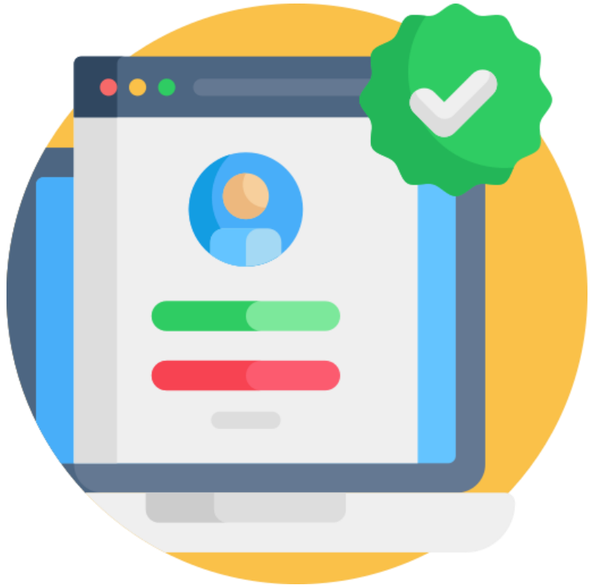
Managed Policy

Managed policies assign the appropriate permissions to individuals, groups, and roles easily compared to user-written policies.



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Amazon Resource Names (ARNs)

- AWS resources are uniquely identified by Amazon Resource Names (ARNs).
- When you need to specify a resource across all of AWS, such as in IAM policies, Amazon Relational Database Service (Amazon RDS) tags, and API calls, you must provide its ARN.

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ARN Format

ARNs follow standard formats, and the specific format for a resource is specified by that resource itself.

```
arn:partition:service:region:account-id:resource-id
arn:partition:service:region:account-id:resource-type/resource-id
arn:partition:service:region:account-id:resource-type:resource-id
```

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AWS Principals

The following principles are used in AWS:

01	AWS account and root user
02	IAM roles
03	Role sessions
04	IAM users

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AWS Principals

The following principles are used in AWS:

05	Federated user sessions
06	AWS services
07	All principals

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IAM Roles

These roles define permissions and policies that govern the access granted to AWS identities.

01 IAM roles operate similarly to IAM users.

02 They lack password protection and do not necessitate access keys.

03 These roles are available for use by any entity requiring them.

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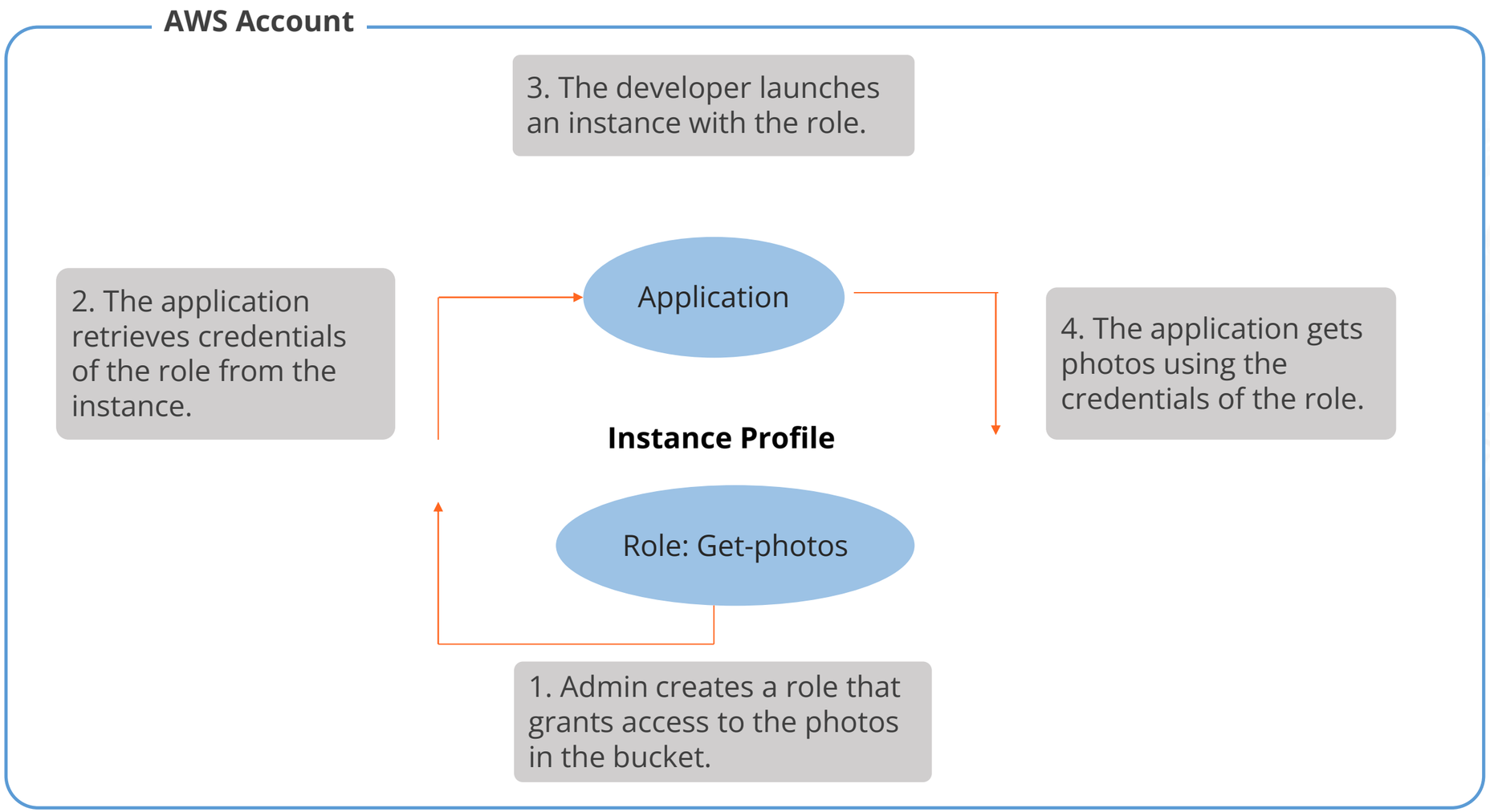
Various Functions of IAM Roles

IAM roles grant access to users, applications, and services that lack permission to utilize AWS resources.



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Creating and Adding Policies to Groups Using Users

Duration:10 min

Problem Statement:

You have been assigned a task to create and add a policy to the group using a user to enable security management in various systems and applications.

Outcome:

This setup ensures that the RDS instance is created with the necessary security and configuration parameters, making it ready for use in your application.

Note: Refer to the demo document for detailed steps:
01_Creating_and_Adding_Policies_to_Groups_Using_Users

ASSISTED PRACTICE

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Assisted Practice: Guidelines

Steps to be followed are:

1. Create and manage policy
2. Attach policy and permissions directly to the group using group users
3. Create and manage S3 versioning

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Configuring Resource-based Policy Using Principals

Duration:10 min

Problem Statement:

You have been assigned a task to configure resource-based policies using principals to enable the access to AWS resources.

Outcome:

This setup ensures that the RDS instance is created with the necessary security and configuration parameters, making it ready for use in your application.

Note: Refer to the demo document for detailed steps: 02_Configuring_Resource-based_Policy_Using_Principals

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Assisted Practice: Guidelines

Steps to be followed are:

1. Create users and attach policies to them
2. Generate the policy using principals

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Quick Check

An AWS administrator wants only users in a specific IAM group, with MFA enabled, to access a particular S3 bucket. Which approach should they take?

A. Create an S3 bucket policy with specific ARNs

B. Attach an IAM policy to the group requiring MFA

C. Enable MFA on the bucket with a resource policy

D. Assign a resource policy to the bucket with MFA.

AWS Cognito

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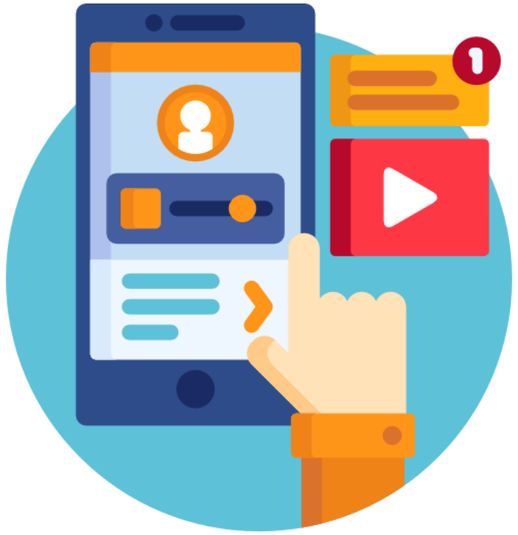
AWS Cognito



Amazon Cognito manages web and mobile app authentication, authorization, and user management.

Users can sign in directly using a username and password or through third-party services like Facebook, Amazon, Google, or Apple.

It comprises two key components: user pools and identity pools.

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Cognito: Example

This example is a combination of an Amazon Cognito user pool and an identity pool.

App users register through a user pool and obtain user pool tokens upon successful authentication.

Subsequently, the user app employs an identity pool to trade user pool tokens for AWS credentials.

With AWS credentials in hand, app users gain access to various AWS services, including Amazon S3 and DynamoDB.

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AWS Control Tower

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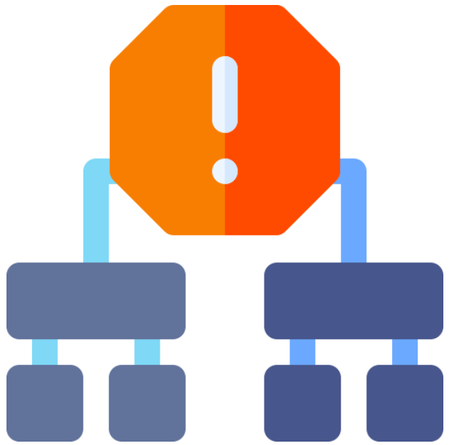
AWS Control Tower



- It is the easiest way to establish and manage a secure, multi-account AWS environment referred to as a landing zone.
- It provides users with a well-structured landing zone equipped with AWS Organizations, ensuring continuous account management, governance, and deployment best practices.

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Benefits of AWS Control Tower

AWS Control Tower offers a range of advantages, including:

- Rapid environment setup
- Automated policy management
- Policy overview

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Cloud Security

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Cloud Security

- In cloud computing, users do not need to manage physical servers or storage components.
- Instead, they opt for software-based security measures to monitor and safeguard the data flowing into and out of their cloud resources.

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Types of Cyber Attacks

The following are some common threats:

- Phishing
- Password management
- Credential leaks
- Network security
- Insider threat
- Security incident recovery planning

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Case Study: Netflix

Location: United States

Industry: On-demand media industry

Challenge

In 2008, Netflix faced a significant challenge when its primary focus was on DVD-by-mail services. During this period, they encountered a database corruption issue that led to a complete halt in DVD shipping for a duration of three days.

Solution

Netflix management decided to shift away from relational systems in their data centers and toward the cloud. The cloud was AWS which allowed the organization to scale as much as it required.

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Amazon VPC

Amazon VPC provides users with full control over their virtual networking environment, encompassing resource allocation, connectivity, and security measures.

Use Cases:

- Launching a simple website or blog
- Hosting multi-tier web applications
- Creating hybrid connections

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Amazon CloudFront

Amazon CloudFront is a CDN solution designed for great performance, security, and developer simplicity.

Use Cases:

- Deliver fast, secure websites
- Accelerate dynamic content delivery and APIs
- Stream live and on-demand video

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Creating and Changing AWS Security Groups

Duration:10 min

Problem Statement:

You have been assigned a task configure AWS security groups to control inbound and outbound traffic to and from AWS resources.

Outcome:

This setup ensures that the RDS instance is created with the necessary security and configuration parameters, making it ready for use in your application.

Note: Refer to the demo document for detailed steps:
03_Creating_and_Changing_AWS_Security_Groups

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Assisted Practice: Guidelines

Steps to be followed are:

1. Create the security group
2. Change the security group

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Quick Check

A company wants to improve the security of its web application hosted on AWS by protecting it from DDoS attacks while ensuring low-latency content delivery to users globally. Which combination of AWS services should they use?

A. Amazon VPC and AWS WAF

B. Amazon CloudFront and AWS Shield

C. Amazon VPC and Amazon CloudFront

D. AWS Shield and AWS Security Groups

AWS Web Application Firewall (WAF)

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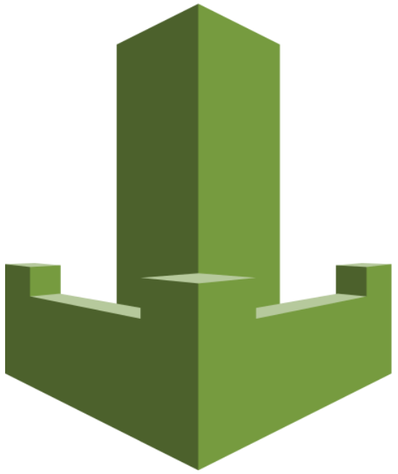
What Is AWS WAF?

It is a web application firewall for monitoring HTTP/HTTPS requests to CloudFront, API Gateway, or Application Load Balancer.



AWS WAF

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AWS WAF

It enables users to:

01	Allow all the requests except the ones that they specify
02	Block all the requests except the ones that they specify
03	Count the requests that match the properties that they specify

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How to Handle False Positives in WAF?

Users must use a web browser. To check for a false positive of **style==xxx** on their **example.com** domain, enter **example.com/style==xxx** in the web browser. The response is error code **403 Forbidden**.

```
$ curl -ikv http://example.com/[false positive]
```

Note:

Users apply curl and replace **users' false positive** for **false positive**.

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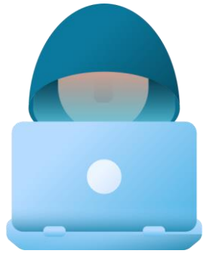
Benefits of AWS WAF



Offers additional protection against web attacks

Provides real-time information and samples of web requests

Offers automated administration

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Creating and Configuring AWS WAF

Duration:10 min

Problem Statement:

You have been assigned a task to configure AWS WAF to protect your web applications and APIs hosted on AWS environment.

Outcome:

This setup ensures that the RDS instance is created with the necessary security and configuration parameters, making it ready for use in your application.

Note: Refer to the demo document for detailed steps:
04_Creating_and_Configuring_AWS_WAF

ASSISTED PRACTICE

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Assisted Practice: Guidelines

Steps to be followed are:

1. Create an IP set
2. Create Web ACL
3. Create a custom rule in Web ACL

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Quick Check

An e-commerce site using AWS WAF is blocking legitimate traffic. How should the security team address the false positives?

A. Disable the rule causing the false positives

B. Create an exception in the rule for the specific traffic

C. Increase the rule's sensitivity to catch fewer threats

D. Remove AWS WAF from the web application.

AWS Shield

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What Is AWS Shield?

AWS Shield is a managed service designed to protect applications hosted on AWS from Distributed Denial of Service (DDoS) attacks.

AWS Shield

This service provides a comprehensive defense mechanism against malicious attempts to overwhelm and disrupt applications, ensuring their availability and performance.

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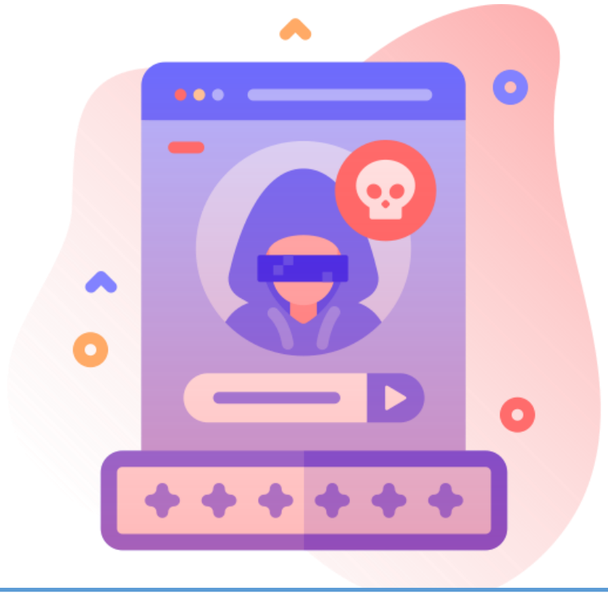
AWS Shield Advance

AWS Shield Advanced is an enhanced subscription service that offers a comprehensive suite of features for advanced protection against Distributed Denial of Service (DDoS) attacks.



This service provides a higher level of security and customization to safeguard applications hosted on AWS.

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Benefits of AWS Shield

AWS Shield offers a range of advantages, including:

01 Seamless integration and deployment

02 Customizable protection

03 Managed protection and attack visibility

04 Cost-efficiency

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AWS Secrets Manager

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AWS Secrets Manager

It is a powerful service that offers a secure and efficient solution for managing sensitive credentials and secrets used in applications and services.

AWS Secrets Manager

This service eliminates the need to hardcode sensitive information, such as passwords and API keys, directly into code or configuration files.

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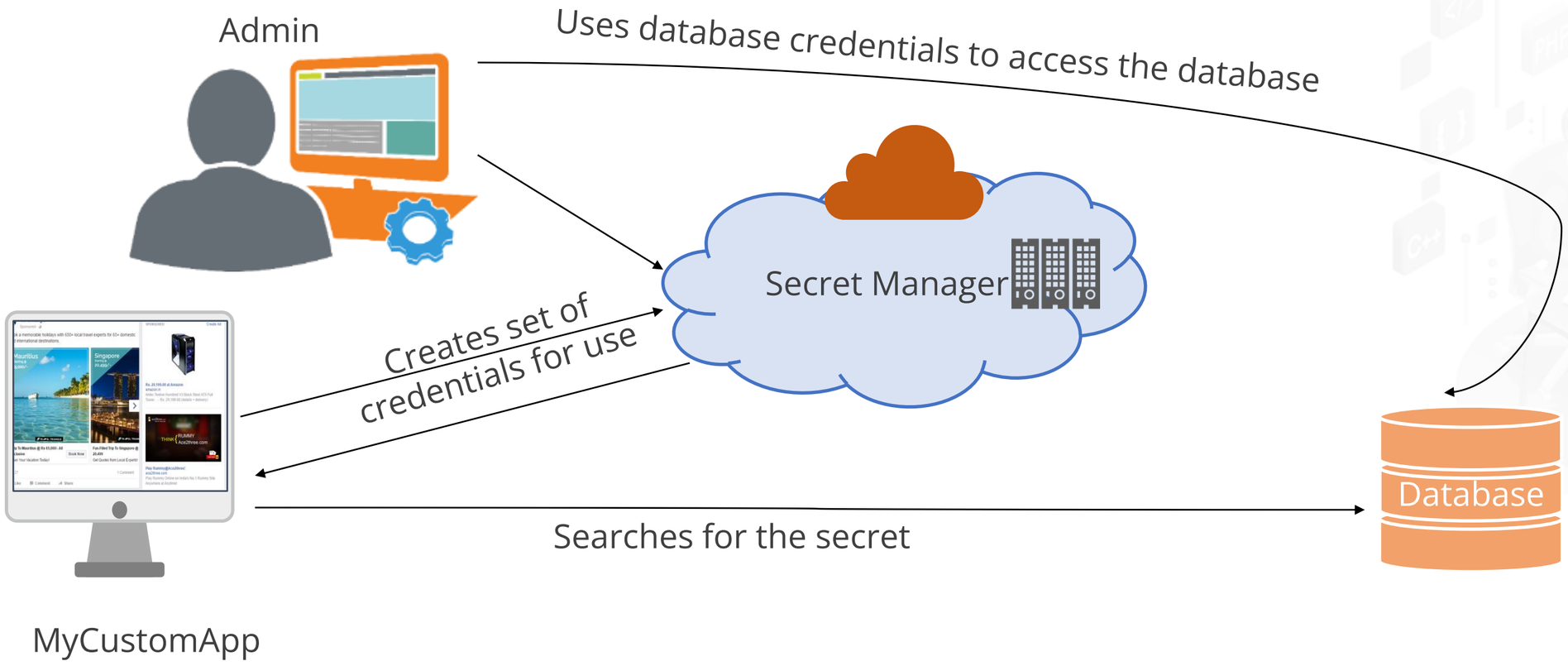
AWS Secrets Manager Scenario

The image below depicts how the users can save database credentials in Secrets Manager and then use those credentials in an application to access the database.



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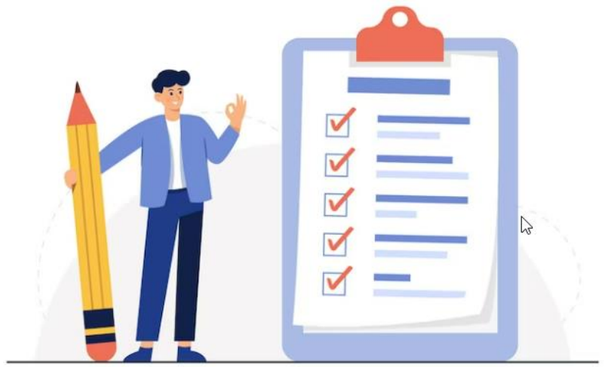
AWS Secrets Manager: Features



- Allows the users to replace stored credentials with a runtime call to the Secrets Manager Web service dynamically
- Is easy to create, leaving users time to focus on creating the applications
- Returns the most recently encrypted secret value version
- Provides users with scalability and reliability provided by AWS

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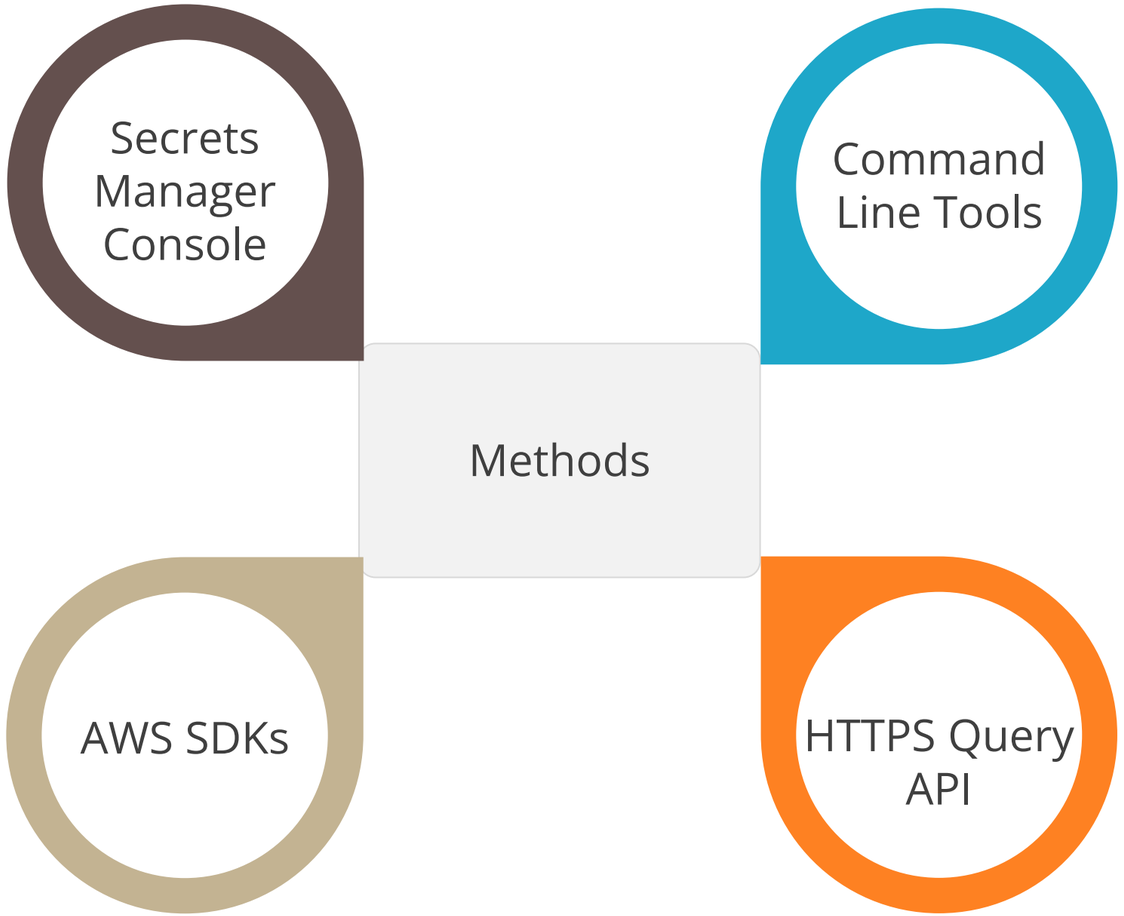
Working with AWS Secrets Manager

Any of the following methods can be used to interact with Secrets Manager:



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Working with AWS Secrets Manager

Secrets Manager Console

The browser-based Secrets Manager console allows users to manage the secrets and perform any task related to the secrets.

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Working with AWS Secrets Manager

Command line tools

The AWS command line tools enable users to perform Secrets Manager and other AWS tasks by issuing commands from the system command line.

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Working with AWS Secrets Manager

AWS SDKs

The AWS SDKs include libraries and sample code for a variety of programming languages and platforms, including Java, Python, Ruby, .NET, and others.

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Working with AWS Secrets Manager

HTTPS Query API

It allows users to access Secrets Manager and AWS programmatically.

API

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Secrets Manager Administrator Permissions

End users should not be granted administrator permissions that help them to carry out the following tasks:

- Focusing on how an organization performs work
- Viewing the value delivery of an organization

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Permissions to Access Secrets

Users may manage which people or services have access to the secrets by using IAM authorization policies. A permissions policy specifies who can do what on which resources.

Users may:

- Attach a permissions policy to an identity
- Attach a permissions policy to a secret

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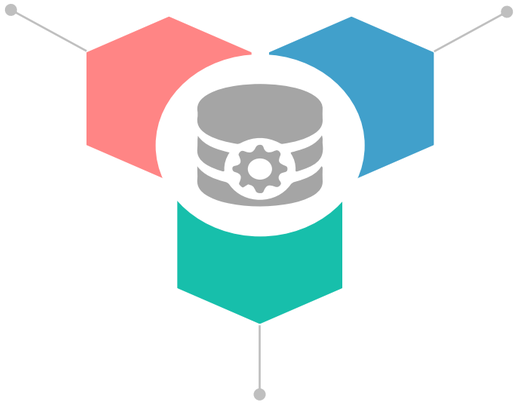
Attach a Permissions Policy to an Identity

Identity-based policies can be used to:

Give a user access to several secrets



Allow an IAM group to access secrets

Control who can create new secrets and who can access already generated secrets

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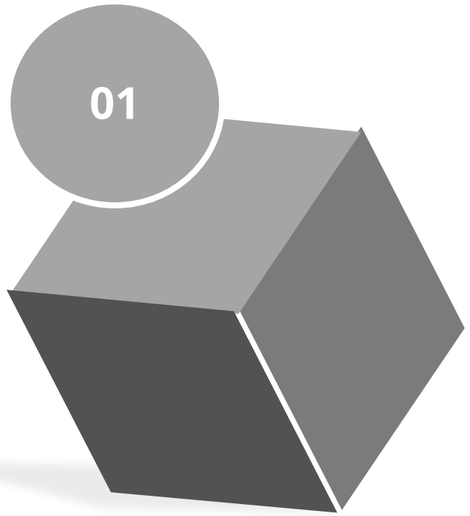
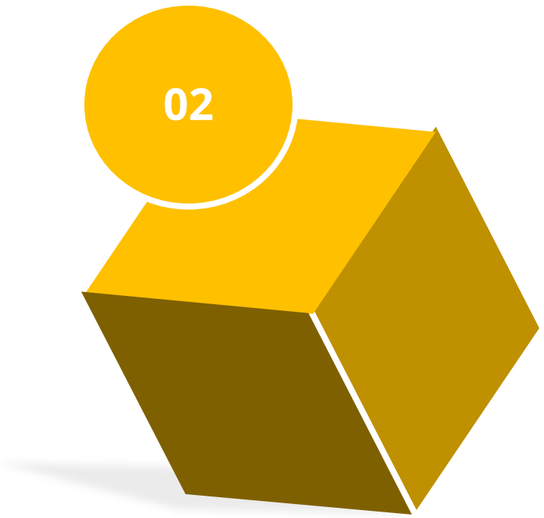
Attach a Permissions Policy to a Secret

In a resource-based policy, users determine who will have access to the secret and what actions they may take with it. Resource-based policies can be used to:



Grant access to a single secret to multiple users and roles



Grant access to users or roles in other AWS accounts

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Quick Check

A developer needs to access a database password stored in AWS Secrets Manager, but the request is denied. What is the most likely reason?

A. The secret does not exist.

B. The developer lacks the required IAM permissions.

C. The database is not connected to Secrets Manager.

D. The secret has expired.

AWS Systems Manager

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AWS Systems Manager

It offers a comprehensive suite of tools designed to aid users in efficiently managing the infrastructure and applications hosted on the AWS Cloud.

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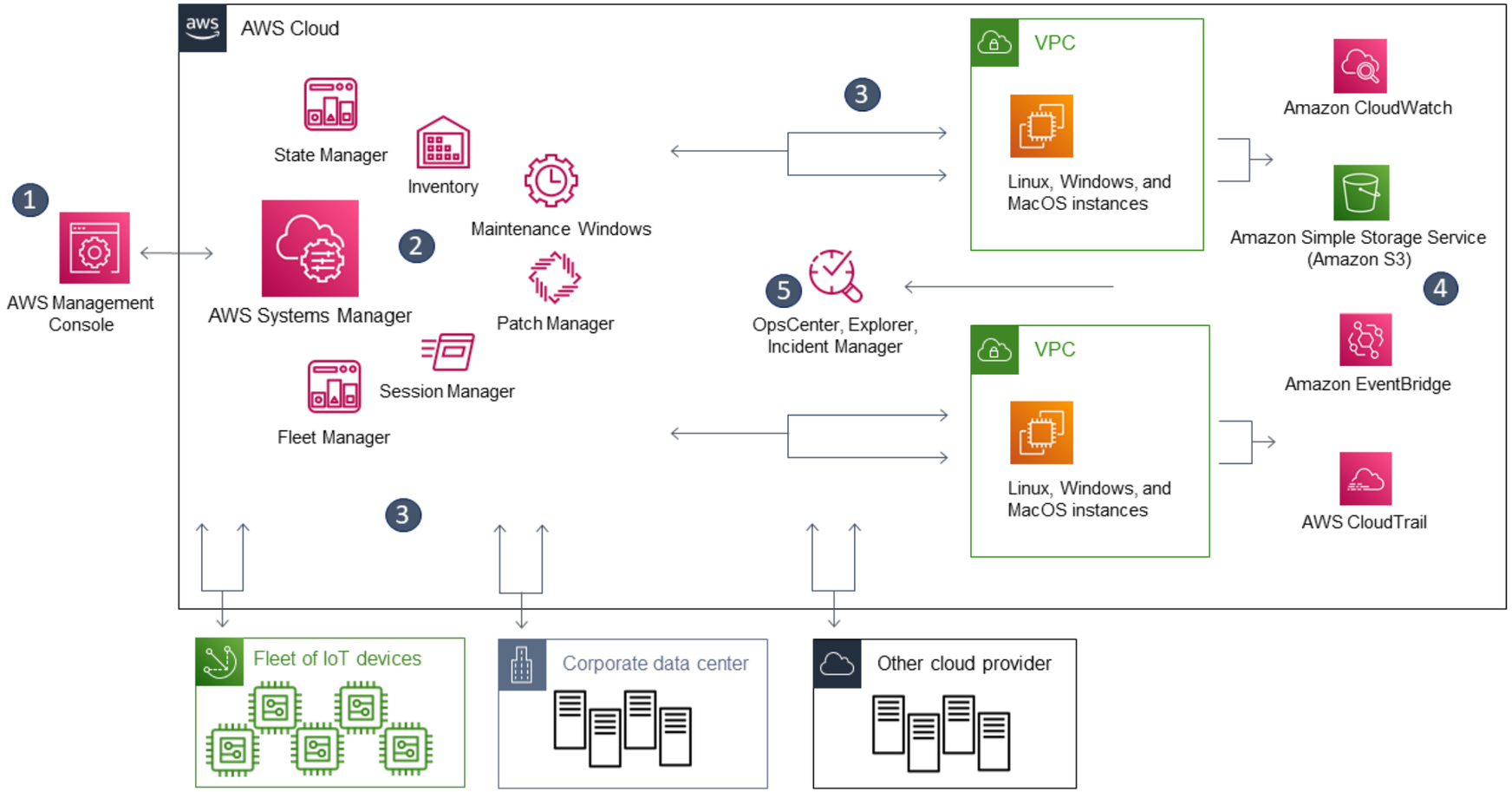
How Systems Manager Works?

The activities performed by the Systems Manager on the resources are shown in the following diagram with a detailed description of each interaction between different components:



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Systems Manager Service Name History

The previous names for AWS Systems Manager are listed below:

- Systems Manager Agent: SSM Agent
- Systems Manager parameters: SSM parameters
- Systems Manager service endpoints: ssm.region.amazonaws.com
- AWS CloudFormation resource types: AWS::SSM::Document

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Systems Manager Service Name History

The previous names for AWS Systems Manager are listed below:

AWS Config rule identifier: EC2_INSTANCE _MANAGED_BY_SSM

AWS Command Line Interface (AWS CLI) commands: aws ssm describe-patch-baselines

AWS Identity and Access Management (IAM) managed policy names: AmazonSSMReadOnlyAccess

Systems Manager resource ARNs: arn:aws:ssm:region:account-id:patchbaseline/pb-07d8884178EXAMPLE

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AWS Config

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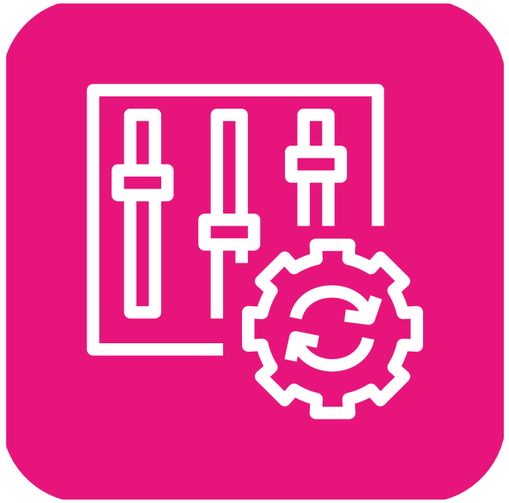
AWS Config

It is a powerful service that provides continuous monitoring and assessment of your AWS resources' configurations.



AWS Config ensures that your infrastructure complies with desired configurations and helps you maintain security, compliance, and governance across your cloud environment.

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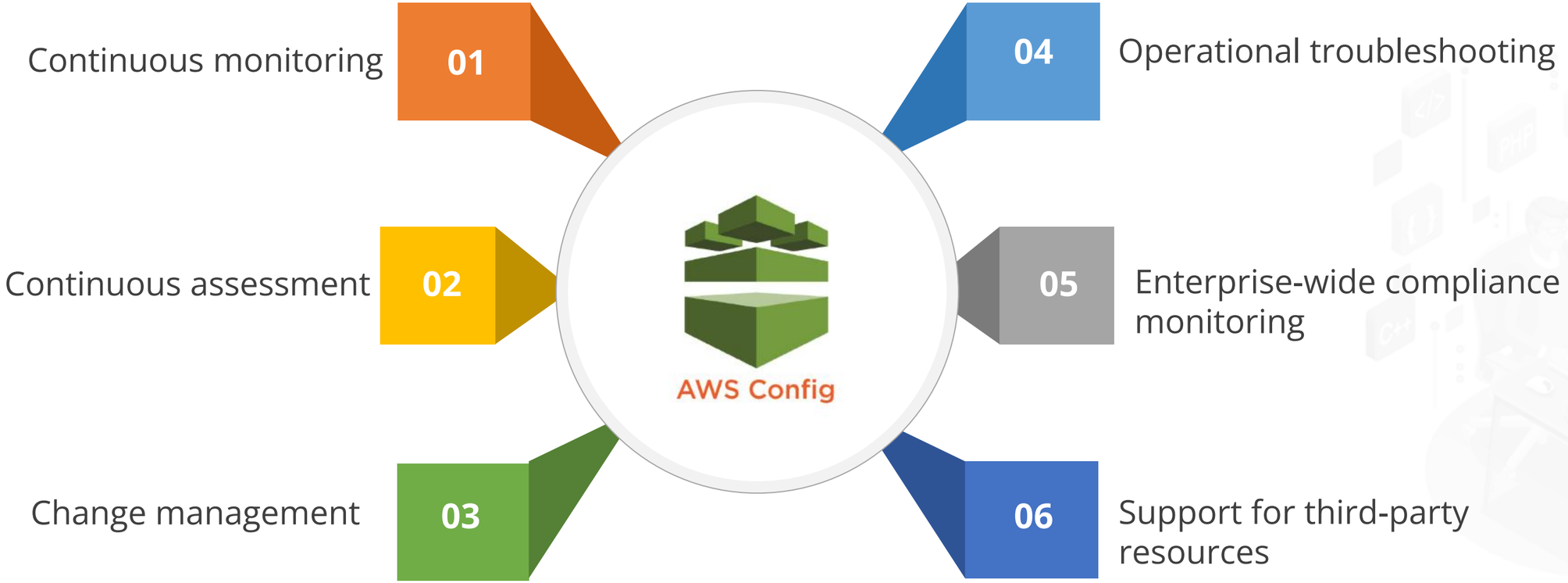
Benefits of AWS Config



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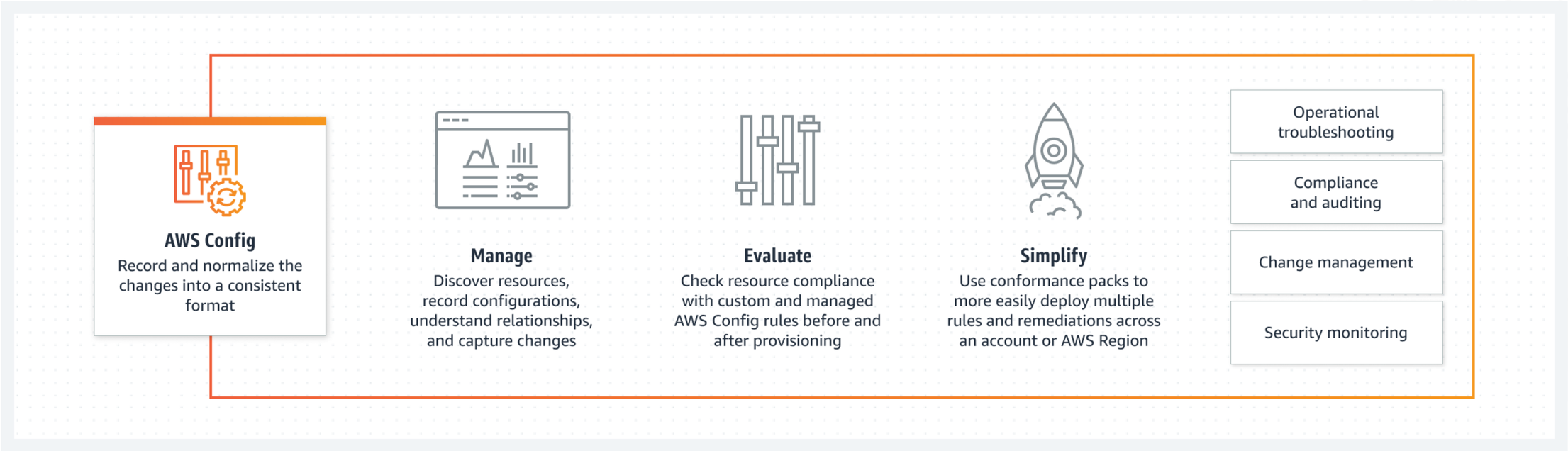
How Does AWS Config Work?

AWS Config generates configuration items when a resource's configuration changes, and it maintains a historical record of these items from the moment the configuration recorder is initiated by users.



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Setting Up AWS Config and Creating Rules in it

Duration:15 min

Problem Statement:

You have been assigned a task to configure AWS Config and create rules in it to enable compliance monitoring in the AWS environment.

Outcome:

This setup ensures that the RDS instance is created with the necessary security and configuration parameters, making it ready for use in your application.

Note: Refer to the demo document for detailed steps:
05_Setting_Up_AWS_Config_and_Creating_Rules_in_it

ASSISTED PRACTICE

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Assisted Practice: Guidelines

Steps to be followed are:

1. Set up AWS Config
2. Create rules in AWS Config

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Quick Check

An administrator needs to ensure that all EC2 instances in their AWS environment comply with specific security configurations. Which AWS service should they use to continuously monitor and enforce these configurations?

A. AWS Systems Manager Patch Manager

B. AWS Config

C. AWS Systems Manager State Manager

D. AWS CloudTrail

Key Takeaways

- IAM policies let users control employee and system access to ensure least-privilege access.
- For web and mobile apps, Amazon Cognito enables quick, secure user authentication, authorization, and management.
- AWS WAF gives control over how traffic enters the applications, allowing users to establish security rules to block common attack vectors.
- AWS Shield offers always-on detection and automatic inline mitigations that reduce application downtime and latency.

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Key Takeaways

- Users can automate the process of creating and configuring numerous accounts made possible by AWS Control Tower.
- Users can secure, analyze, and manage secrets in the AWS Cloud, on external services, and on-premises using Secrets Manager.
- Users can compare their environment to security best practices and standards with the aid of AWS Security Hub.

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Create and Configure Groups and Users Using Policies

Duration: 30 mins

Project agenda: To demonstrate how to use AWS IAM to create and configure groups and users with different policies and permissions

Description: The admin of a corporation wants to create three groups of IAM users, each with two users and two policies assigned to them. The policies will define the permissions and roles for the users to perform their tasks in their respective groups.

Perform the following:

1. Create a group
2. Configure the user for CLI operation

LESSON-END PROJECT

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Thank You

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